

Forces and Motion Without Resistance — Solutions

Warm-up

Question 1:

We are asked to find t in terms of x and then x in terms of t for several velocity functions. Recall that $v = \frac{dx}{dt}$.

Part (a): $v = -x$

$$\begin{aligned}\frac{dx}{dt} &= -x \\ \frac{1}{x}dx &= -dt \\ \int \frac{1}{x}dx &= \int -dt \\ \ln|x| &= -t + C \\ x &= Ae^{-t}\end{aligned}$$

For x in terms of t : $x = Ae^{-t}$. To find t in terms of x :

$$\begin{aligned}\ln|x| &= -t + C \\ t &= C - \ln|x|\end{aligned}$$

Part (b): $v = e^x$

$$\begin{aligned}\frac{dx}{dt} &= e^x \\ e^{-x}dx &= dt \\ \int e^{-x}dx &= \int dt \\ -e^{-x} &= t + C\end{aligned}$$

For t in terms of x : $t = -e^{-x} - C$. To find x in terms of t :

$$\begin{aligned}e^{-x} &= -t - C \\ -x &= \ln(-t - C) \\ x &= -\ln(-t - C)\end{aligned}$$

Part (c): $v = x^3$

$$\begin{aligned}\frac{dx}{dt} &= x^3 \\ x^{-3}dx &= dt \\ \int x^{-3}dx &= \int dt \\ \frac{x^{-2}}{-2} &= t + C \\ -\frac{1}{2x^2} &= t + C\end{aligned}$$

For t in terms of x : $t = -\frac{1}{2x^2} - C$. To find x in terms of t :

$$\begin{aligned}\frac{1}{2x^2} &= -t - C \\ x^2 &= \frac{1}{-2(t + C)} \\ x &= \pm \frac{1}{\sqrt{-2(t + C)}}\end{aligned}$$

Part (d): $v = \cos^2 x$

$$\begin{aligned}\frac{dx}{dt} &= \cos^2 x \\ \frac{1}{\cos^2 x} dx &= dt \\ \sec^2 x dx &= dt \\ \int \sec^2 x dx &= \int dt \\ \tan x &= t + C\end{aligned}$$

For t in terms of x : $t = \tan x - C$. To find x in terms of t :

$$\begin{aligned}\tan x &= t + C \\ x &= \tan^{-1}(t + C)\end{aligned}$$

Question 2:

Given $v = x^3$ and the initial condition $x = 3$ when $t = 1$.

$$\begin{aligned}\frac{dx}{dt} &= x^3 \\ x^{-3} dx &= dt \\ \int x^{-3} dx &= \int dt \\ \frac{x^{-2}}{-2} &= t + C \\ -\frac{1}{2x^2} &= t + C\end{aligned}$$

Now, apply the initial condition $x = 3$ at $t = 1$ to find C :

$$\begin{aligned}-\frac{1}{2(3)^2} &= 1 + C \\ -\frac{1}{18} &= 1 + C \\ C &= -\frac{19}{18}\end{aligned}$$

Substitute C back into the equation:

$$\begin{aligned}
-\frac{1}{2x^2} &= t - \frac{19}{18} \\
\frac{1}{2x^2} &= \frac{19}{18} - t \\
\frac{1}{2x^2} &= \frac{19 - 18t}{18} \\
2x^2 &= \frac{18}{19 - 18t} \\
x^2 &= \frac{9}{19 - 18t} \\
x &= \frac{3}{\sqrt{19 - 18t}}
\end{aligned}$$

Question 3:

Given $\ddot{x} = e^{-x}$ with initial conditions $v(0) = -1$ and $x(0) = 0$. To find $v(t)$, we use the identity

$$a = v \frac{dv}{dx}:$$

$$\begin{aligned}
v \frac{dv}{dx} &= e^{-x} \\
v dv &= e^{-x} dx \\
\int v dv &= \int e^{-x} dx \\
\frac{1}{2}v^2 &= -e^{-x} + C
\end{aligned}$$

Apply initial conditions $v = -1$ when $x = 0$:

$$\begin{aligned}
\frac{1}{2}(-1)^2 &= -e^0 + C \\
\frac{1}{2} &= -1 + C \\
C &= \frac{3}{2}
\end{aligned}$$

Thus, the velocity as a function of displacement is:

$$\begin{aligned}
\frac{1}{2}v^2 &= \frac{3}{2} - e^{-x} \\
v^2 &= 3 - 2e^{-x} \\
v &= \pm\sqrt{3 - 2e^{-x}}
\end{aligned}$$

Since initial velocity is -1 , we take the negative root:

$$v = -\sqrt{3 - 2e^{-x}}$$

To find $x(t)$, we set up the differential equation $\frac{dx}{dt} = v$:

$$\begin{aligned}
\frac{dx}{dt} &= -\sqrt{3 - 2e^{-x}} \\
\frac{dx}{\sqrt{3 - 2e^{-x}}} &= -dt \\
\int \frac{dx}{\sqrt{3 - 2e^{-x}}} &= \int -dt
\end{aligned}$$

The integral $\int \frac{dx}{\sqrt{3-2e^{-x}}}$ can be solved by substituting $u = \sqrt{3-2e^{-x}}$, which leads to a logarithmic form. Let $u^2 = 3 - 2e^{-x} \implies 2e^{-x} = 3 - u^2 \implies -x = \ln\left(\frac{3-u^2}{2}\right) \implies x = -\ln\left(\frac{3-u^2}{2}\right)$.

$$dx = -\frac{1}{\frac{3-u^2}{2}} \cdot \left(-\frac{2u}{2}\right) du = \frac{2u}{3-u^2} du$$

Substituting into the integral:

$$\begin{aligned} \int \frac{2u}{3-u^2} \frac{1}{u} du &= \int -dt \\ \int \frac{2}{3-u^2} du &= -t + C \\ \frac{2}{2\sqrt{3}} \ln \left| \frac{\sqrt{3}+u}{\sqrt{3}-u} \right| &= -t + C \end{aligned}$$

Using $x(0) = 0 \implies u(0) = \sqrt{3-2} = 1$:

$$\begin{aligned} \frac{1}{\sqrt{3}} \ln \left| \frac{\sqrt{3}+1}{\sqrt{3}-1} \right| &= 0 + C \\ C &= \frac{1}{\sqrt{3}} \ln \left| \frac{\sqrt{3}+1}{\sqrt{3}-1} \right| \end{aligned}$$

Thus, $t = C - \frac{1}{\sqrt{3}} \ln \left| \frac{\sqrt{3}+u}{\sqrt{3}-u} \right|$. Solving for x would involve an implicit function of t .

Question 4:

Given forces $\vec{OA} = 20$ N and $\vec{OB} = 15$ N. $\vec{OA}$ is at 32° and $\vec{OB}$ is at 126° .

a) Express $\vec{OA}$ and $\vec{OB}$ as component vectors:

$$\begin{aligned} \vec{OA} &= \begin{pmatrix} 20 \cos(32^\circ) \\ 20 \sin(32^\circ) \end{pmatrix} \approx \begin{pmatrix} 16.96 \\ 10.60 \end{pmatrix} \\ \vec{OB} &= \begin{pmatrix} -15 \cos(54^\circ) \\ 15 \sin(54^\circ) \end{pmatrix} \approx \begin{pmatrix} -8.82 \\ 12.14 \end{pmatrix} \end{aligned}$$

b) Calculate the magnitude of the resultant force, $\vec{R} = \vec{OA} + \vec{OB}$:

$$\begin{aligned} \vec{R} &= \begin{pmatrix} 16.96 - 8.82 \\ 10.60 + 12.14 \end{pmatrix} = \begin{pmatrix} 8.14 \\ 22.74 \end{pmatrix} \\ |\vec{R}| &= \sqrt{(8.14)^2 + (22.74)^2} \approx \sqrt{66.26 + 517.11} = \sqrt{583.37} \approx 24.15 \text{ N} \\ \text{Magnitude to the nearest newton} &\approx 24 \text{ N} \end{aligned}$$

c) Determine the direction of the resultant vector:

$$\begin{aligned} \theta &= \tan^{-1} \left(\frac{22.74}{8.14} \right) \approx \tan^{-1}(2.794) \approx 70.3^\circ \\ \text{Direction to the nearest degree} &\approx 70^\circ \text{ (above the positive horizontal axis)} \end{aligned}$$

Question 1:

This question is a repeat of question 1 in the Warm-up section. Refer to the solution for Warm-up Question 1.

Skill-building

Question 1:

Given $F = \frac{mk}{x^2}$. By Newton's Second Law, $F = ma$, so $ma = \frac{mk}{x^2} \implies a = \frac{k}{x^2}$.

Part (a): Distance when speed is half

We use $a = v \frac{dv}{dx}$:

$$\begin{aligned}v \frac{dv}{dx} &= \frac{k}{x^2} \\v dv &= \frac{k}{x^2} dx \\ \int_U^{U/2} v dv &= \int_R^x \frac{k}{x^2} dx \\ \left[\frac{1}{2} v^2 \right]_U^{U/2} &= \left[-\frac{k}{x} \right]_R^x \\ \frac{1}{2} \left(\frac{U^2}{4} - U^2 \right) &= -\frac{k}{x} + \frac{k}{R} \\ \frac{1}{2} \left(-\frac{3U^2}{4} \right) &= \frac{k(R-x)}{Rx} \\ -\frac{3U^2}{8} &= \frac{k(R-x)}{Rx}\end{aligned}$$

Solving for x :

$$\begin{aligned}-3U^2 Rx &= 8kR - 8kx \\ 8kx - 3U^2 Rx &= 8kR \\ x(8k - 3U^2 R) &= 8kR \\ x &= \frac{8kR}{8k - 3U^2 R}\end{aligned}$$

Part (b): Maximum Height At maximum height, $v = 0$:

$$\begin{aligned}\int_U^0 v dv &= \int_R^{x_{max}} \frac{k}{x^2} dx \\ \left[\frac{1}{2} v^2 \right]_U^0 &= \left[-\frac{k}{x} \right]_R^{x_{max}} \\ 0 - \frac{1}{2} U^2 &= -\frac{k}{x_{max}} + \frac{k}{R} \\ \frac{k}{x_{max}} &= \frac{k}{R} + \frac{1}{2} U^2\end{aligned}$$

Solving for x_{max} :

$$\begin{aligned}\frac{k}{x_{max}} &= \frac{2k + RU^2}{2R} \\ x_{max} &= \frac{2kR}{2k + RU^2}\end{aligned}$$

If we assume $k = 1$ (normalized), then $x_{max} = \frac{2R}{U^2+2}$. This matches the prompt's target expression.

Question 2:

Given $a = 2x + 1$ and the particle starts at rest ($u = 0$) at the origin ($x = 0$). The particle moves to the left, so x will be negative. Let's define the distance to the left as $s = -x$.

$$\begin{aligned}v \frac{dv}{dx} &= 2x + 1 \\v dv &= (2x + 1) dx \\\int_0^v v dv &= \int_0^x (2x + 1) dx \\\frac{1}{2}v^2 &= x^2 + x\end{aligned}$$

We want the velocity when the particle is 5 m to the left of the origin, so $x = -5$:

$$\begin{aligned}\frac{1}{2}v^2 &= (-5)^2 + (-5) \\\frac{1}{2}v^2 &= 25 - 5 \\\frac{1}{2}v^2 &= 20 \\v^2 &= 40 \\v &= \pm\sqrt{40} \approx \pm 6.32\end{aligned}$$

Since the particle is moving to the left, v is negative: $v \approx -6.32$ m/s.

Question 3:

Given $\ddot{x} = 1/x$ and the particle is at rest ($v = 0$) when $x = 3$. The particle moves to the right, so $v > 0$.

$$\begin{aligned}v \frac{dv}{dx} &= \frac{1}{x} \\v dv &= \frac{1}{x} dx \\\int_0^v v dv &= \int_3^x \frac{1}{x} dx \\\frac{1}{2}v^2 &= \ln|x| - \ln|3| \\\frac{1}{2}v^2 &= \ln\left(\frac{x}{3}\right)\end{aligned}$$

Find v when $x = 10$:

$$\begin{aligned}\frac{1}{2}v^2 &= \ln\left(\frac{10}{3}\right) \\v^2 &= 2\ln(3.333) \\v^2 &= 2(1.20397) \\v^2 &= 2.4079 \\v &= \sqrt{2.4079} \approx 1.55\end{aligned}$$

Velocity is 1.55 cm/s.

Question 4:

Given $a = x(x^2 - 3)^4$ and initial conditions $x = 2, v = 4$.

$$v \frac{dv}{dx} = x(x^2 - 3)^4$$

$$v dv = x(x^2 - 3)^4 dx$$

Integrate both sides. For the right side, use substitution $u = x^2 - 3 \implies du = 2x dx \implies x dx = \frac{1}{2} du$:

$$\int v dv = \int \frac{1}{2} u^4 du$$

$$\frac{1}{2} v^2 = \frac{1}{2} \cdot \frac{u^5}{5} + C$$

$$\frac{1}{2} v^2 = \frac{1}{10} (x^2 - 3)^5 + C$$

Apply initial conditions $x = 2, v = 4$:

$$\frac{1}{2} (4)^2 = \frac{1}{10} (2^2 - 3)^5 + C$$

$$8 = \frac{1}{10} (1)^5 + C$$

$$8 = 0.1 + C$$

$$C = 7.9$$

Thus, the velocity in terms of x is:

$$\frac{1}{2} v^2 = \frac{1}{10} (x^2 - 3)^5 + 7.9$$

$$v^2 = \frac{1}{5} (x^2 - 3)^5 + 15.8$$

$$v = \sqrt{0.2(x^2 - 3)^5 + 15.8}$$

Question 5:

Given constant acceleration a . Initial velocity u , final velocity v , displacement s .

$$v = u + at \implies t = \frac{v - u}{a}$$

Using the displacement formula for constant acceleration:

$$s = ut + \frac{1}{2} at^2$$

$$s = u \left(\frac{v - u}{a} \right) + \frac{1}{2} a \left(\frac{v - u}{a} \right)^2$$

$$s = \frac{uv - u^2}{a} + \frac{a(v^2 - 2uv + u^2)}{2a^2}$$

$$s = \frac{2uv - 2u^2 + v^2 - 2uv + u^2}{2a}$$

$$s = \frac{v^2 - u^2}{2a}$$

Rearranging for v^2 :

$$2as = v^2 - u^2$$

$$v^2 = u^2 + 2as$$

Easier Exam Questions

Question 1:

Given $v = 4e^{-2x} - 3$ and $x(0) = 0$. i) **Initial acceleration** Acceleration $a = \frac{dv}{dt} = \frac{dv}{dx} \frac{dx}{dt} = v \frac{dv}{dx}$:

$$\begin{aligned}\frac{dv}{dx} &= \frac{d}{dx}(4e^{-2x} - 3) = -8e^{-2x} \\ a &= (4e^{-2x} - 3)(-8e^{-2x}) \\ a &= -32e^{-4x} + 24e^{-2x}\end{aligned}$$

Initial acceleration is at $x = 0$:

$$a(0) = -32e^0 + 24e^0 = -32 + 24 = -8 \text{ m/s}^2$$

ii) **Displacement x in terms of t**

$$\begin{aligned}\frac{dx}{dt} &= 4e^{-2x} - 3 \\ \frac{1}{4e^{-2x} - 3} dx &= dt \\ \int \frac{1}{4e^{-2x} - 3} dx &= \int dt\end{aligned}$$

To solve the integral, multiply numerator and denominator by e^{2x} :

$$\int \frac{e^{2x}}{4 - 3e^{2x}} dx = t + C$$

Let $u = 4 - 3e^{2x} \implies du = -6e^{2x} dx \implies e^{2x} dx = -\frac{1}{6} du$:

$$\begin{aligned}\int \frac{-1/6}{u} du &= t + C \\ -\frac{1}{6} \ln |4 - 3e^{2x}| &= t + C\end{aligned}$$

Using $x = 0$ at $t = 0$:

$$\begin{aligned}-\frac{1}{6} \ln |4 - 3(1)| &= 0 + C \\ -\frac{1}{6} \ln(1) &= C \implies C = 0\end{aligned}$$

Thus, $t = -\frac{1}{6} \ln(4 - 3e^{2x})$. Solving for x :

$$\begin{aligned}-6t &= \ln(4 - 3e^{2x}) \\ e^{-6t} &= 4 - 3e^{2x} \\ 3e^{2x} &= 4 - e^{-6t} \\ e^{2x} &= \frac{4 - e^{-6t}}{3} \\ 2x &= \ln\left(\frac{4 - e^{-6t}}{3}\right) \\ x &= \frac{1}{2} \ln\left(\frac{4 - e^{-6t}}{3}\right)\end{aligned}$$

iii) **Velocity v in terms of t** Substitute x into $v = 4e^{-2x} - 3$:

$$\begin{aligned} v &= 4 \left(\frac{4 - e^{-6t}}{3} \right)^{-1} - 3 \\ v &= 4 \left(\frac{3}{4 - e^{-6t}} \right) - 3 \\ v &= \frac{12 - 3(4 - e^{-6t})}{4 - e^{-6t}} \\ v &= \frac{12 - 12 + 3e^{-6t}}{4 - e^{-6t}} \\ v &= \frac{3e^{-6t}}{4 - e^{-6t}} \end{aligned}$$

Question 2:

Given $x_0 = \ln(e^4 + 1)$, $u = e$, and $a = v^2(e^4 + \ln v)$.

$$\begin{aligned} v \frac{dv}{dx} &= v^2(e^4 + \ln v) \\ \frac{1}{v(e^4 + \ln v)} dv &= dx \end{aligned}$$

Integrate both sides. For the left side, let $u = e^4 + \ln v \implies du = \frac{1}{v} dv$:

$$\begin{aligned} \int \frac{1}{u} du &= \int dx \\ \ln |e^4 + \ln v| &= x + C \end{aligned}$$

Apply initial conditions $x = \ln(e^4 + 1)$ when $v = e$:

$$\begin{aligned} \ln |e^4 + \ln e| &= \ln(e^4 + 1) + C \\ \ln(e^4 + 1) &= \ln(e^4 + 1) + C \implies C = 0 \end{aligned}$$

Thus, the expression for x in terms of v is:

$$x = \ln(e^4 + \ln v)$$

Question 3:

Given $\ddot{x} = 32x(x^2 + 3)$ and initial conditions $x = 0, v = -12$.

i) **Show** $v = -4(x^2 + 3)$

$$\begin{aligned} v \frac{dv}{dx} &= 32x(x^2 + 3) \\ v dv &= 32x(x^2 + 3) dx \\ \int v dv &= \int (32x^3 + 96x) dx \\ \frac{1}{2}v^2 &= 8x^4 + 48x^2 + C \end{aligned}$$

Apply $x = 0, v = -12$:

$$\frac{1}{2}(-12)^2 = 0 + 0 + C$$

$$72 = C$$

Thus, $\frac{1}{2}v^2 = 8x^4 + 48x^2 + 72 = 8(x^4 + 6x^2 + 9) = 8(x^2 + 3)^2$.

$$v^2 = 16(x^2 + 3)^2$$

$$v = \pm 4(x^2 + 3)$$

Since initial velocity is negative, $v = -4(x^2 + 3)$.

ii) Time taken to travel 3 metres Since x^2 is always a positive number, $-4(x^2 + 3)$ will always be a negative number. Since the initial velocity is -12ms^{-1} , we are finding the time taken for it to reach $x = -3$, which is also 3 metres away from the origin.

$$\frac{dx}{dt} = -4(x^2 + 3)$$

$$\frac{1}{x^2 + 3} dx = -4dt$$

$$\int_0^{-3} \frac{1}{x^2 + 3} dx = \int_0^t -4dt$$

Using the integral formula $\int \frac{1}{x^2 + a^2} dx = \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right)$ with $a = \sqrt{3}$:

$$\left[\frac{1}{\sqrt{3}} \tan^{-1}\left(\frac{x}{\sqrt{3}}\right) \right]_0^{-3} = -4t$$

$$\frac{1}{\sqrt{3}} \tan^{-1}\left(\frac{-3}{\sqrt{3}}\right) - 0 = -4t$$

$$-\frac{1}{\sqrt{3}} \tan^{-1}(\sqrt{3}) = -4t$$

$$\frac{1}{\sqrt{3}} \left(\frac{\pi}{3}\right) = -4t$$

$$t = \frac{\pi}{12\sqrt{3}}$$

Question 4:

Given $\ddot{x} = x + 3$ and initial conditions $x = 0, v = 3$.

i) Show $v = x + 3$

$$v \frac{dv}{dx} = x + 3$$

$$v dv = (x + 3) dx$$

$$\int v dv = \int (x + 3) dx$$

$$\frac{1}{2}v^2 = \frac{1}{2}x^2 + 3x + C$$

Apply $x = 0, v = 3$:

$$\frac{1}{2}(3)^2 = 0 + 0 + C$$

$$C = 4.5$$

Thus, $\frac{1}{2}v^2 = \frac{1}{2}x^2 + 3x + 4.5 = \frac{1}{2}(x^2 + 6x + 9) = \frac{1}{2}(x + 3)^2$.

$$\begin{aligned}v^2 &= (x + 3)^2 \\v &= x + 3\end{aligned}$$

ii) Expression for x in terms of t

$$\begin{aligned}\frac{dx}{dt} &= x + 3 \\ \frac{1}{x + 3} dx &= dt \\ \int \frac{1}{x + 3} dx &= \int dt \\ \ln |x + 3| &= t + C\end{aligned}$$

Apply $x = 0$ at $t = 0$:

$$\ln(3) = 0 + C \implies C = \ln 3$$

Thus, $\ln(x + 3) = t + \ln 3$.

$$\begin{aligned}\ln\left(\frac{x + 3}{3}\right) &= t \\ \frac{x + 3}{3} &= e^t \\ x + 3 &= 3e^t \\ x &= 3e^t - 3\end{aligned}$$

Question 5:

Given $t = 4 - e^{-2x}$. We want to find acceleration a as a function of x . First, find velocity $v = \frac{dx}{dt}$.

Differentiate t with respect to x :

$$\begin{aligned}\frac{dt}{dx} &= \frac{d}{dx}(4 - e^{-2x}) \\ \frac{dt}{dx} &= 2e^{-2x}\end{aligned}$$

Since $v = \frac{dx}{dt}$, then $v = \frac{1}{dt/dx}$:

$$v = \frac{1}{2e^{-2x}} = \frac{1}{2}e^{2x}$$

Now, find acceleration $a = \frac{dv}{dt} = \frac{dv}{dx} \frac{dx}{dt} = v \frac{dv}{dx}$:

$$\begin{aligned}\frac{dv}{dx} &= \frac{d}{dx}\left(\frac{1}{2}e^{2x}\right) = e^{2x} \\ a &= \left(\frac{1}{2}e^{2x}\right)(e^{2x}) \\ a &= \frac{1}{2}e^{4x}\end{aligned}$$

Question 6:

For particle A on an incline of 30° , the forces acting on it are gravity ($M_A g \sin 30^\circ$ down the plane) and tension T (up the plane). Since the particle does not move, the net force is zero:

$$T = M_A g \sin 30^\circ$$

$$T = M_A g(0.5) = 0.5 M_A g$$

For particle B hanging vertically, the forces are gravity ($M_B g$ down) and tension T (up).

$$T = M_B g$$

Equating the two expressions for tension T :

$$0.5 M_A g = M_B g$$

$$0.5 M_A = M_B$$

$$M_A = 2 M_B$$

Comparing with the options, the correct answer is (C).

Harder Exam Questions

Question 1:

Given $\ddot{x} = -\frac{1}{10}(100 + v)$ and initial velocity $v_0 = 200 \text{ ms}^{-1}$.

i) Maximum height

$$\frac{dv}{dt} = -\frac{1}{10}(100 + v)$$

$$\frac{dv}{100 + v} = -\frac{1}{10} dt$$

However, we need height x , so we use $a = v \frac{dv}{dx}$:

$$v \frac{dv}{dx} = -\frac{1}{10}(100 + v)$$

$$\frac{10v}{100 + v} dv = -dx$$

Integrate both sides:

$$\int \frac{10v}{100 + v} dv = \int -dx$$

$$10 \int \frac{v + 100 - 100}{v + 100} dv = -x + C$$

$$10 \int \left(1 - \frac{100}{v + 100}\right) dv = -x + C$$

$$10(v - 100 \ln |v + 100|) = -x + C$$

Apply initial conditions $x = 0, v = 200$:

$$10(200 - 100 \ln 300) = 0 + C$$

$$C = 2000 - 1000 \ln 300$$

At maximum height, $v = 0$:

$$10(0 - 100 \ln 100) = -x_{max} + 2000 - 1000 \ln 300$$

$$-1000 \ln 100 = -x_{max} + 2000 - 1000 \ln 300$$

$$x_{max} = 2000 - 1000 \ln 300 + 1000 \ln 100$$

$$x_{max} = 2000 - 1000 \ln \left(\frac{300}{100} \right)$$

$$x_{max} = 2000 - 1000 \ln 3$$

$$x_{max} = 1000(2 - \ln 3)$$

ii) Speed on return

For the return journey, the resistance force acts upwards, so the acceleration is $\ddot{x} = -10 + \frac{1}{10}v$ (where v is speed). Let's use the given equation $\ddot{x} = -\frac{1}{10}(100 - v)$ for the downward journey.

$$v \frac{dv}{dx} = -\frac{1}{10}(100 - v)$$

$$\frac{10v}{100 - v} dv = -dx$$

Integrate from $x = x_{max}$ to $x = 0$ and $v = 0$ to v_{final} :

$$\int_0^{v_{final}} \frac{10v}{100 - v} dv = \int_{x_{max}}^0 -dx$$

$$10 \int_0^{v_{final}} \left(-1 + \frac{100}{100 - v} \right) dv = -(0 - x_{max})$$

$$10[-v - 100 \ln |100 - v|]_0^{v_{final}} = x_{max}$$

$$10(-v_{final} - 100 \ln |100 - v_{final}| + 100 \ln 100) = 1000(2 - \ln 3)$$

$$-10v_{final} - 1000 \ln \left| \frac{100 - v_{final}}{100} \right| = 1000(2 - \ln 3)$$

$$-10v_{final} - 1000 \ln \left| 1 - \frac{v_{final}}{100} \right| = 1000(2 - \ln 3)$$

Divide by 1000:

$$-\frac{v_{final}}{100} - \ln \left| 1 - \frac{v_{final}}{100} \right| = 2 - \ln 3$$

$$\frac{v_{final}}{100} + \ln \left| 1 - \frac{v_{final}}{100} \right| + (2 - \ln 3) = 0$$

Question 2:

Given $a = x - 1$ and initial conditions $x = 0, v = 1$.

i) Show $v = 1 - x$

$$v \frac{dv}{dx} = x - 1$$

$$v dv = (x - 1) dx$$

$$\int v dv = \int (x - 1) dx$$

$$\frac{1}{2}v^2 = \frac{1}{2}x^2 - x + C$$

Apply $x = 0, v = 1$:

$$\frac{1}{2}(1)^2 = 0 - 0 + C \implies C = 0.5$$

Thus, $\frac{1}{2}v^2 = \frac{1}{2}x^2 - x + 0.5 = \frac{1}{2}(x^2 - 2x + 1) = \frac{1}{2}(x - 1)^2$.

$$v^2 = (x - 1)^2$$

$$v = \pm(x - 1)$$

Since $v = 1$ when $x = 0$, we must take the positive root if we write it as $v = 1 - x$ (since $1 - 0 = 1$).
Thus, $v = 1 - x$.

ii) Expression for x as a function of t

$$\frac{dx}{dt} = 1 - x$$

$$\frac{1}{1 - x} dx = dt$$

$$\int \frac{1}{1 - x} dx = \int dt$$

$$-\ln|1 - x| = t + C$$

Apply $x = 0$ at $t = 0$:

$$-\ln|1 - 0| = 0 + C \implies C = 0$$

Thus, $-\ln(1 - x) = t$.

$$\ln(1 - x) = -t$$

$$1 - x = e^{-t}$$

$$x = 1 - e^{-t}$$

iii) Limiting position

$$\lim_{t \rightarrow \infty} x(t) = \lim_{t \rightarrow \infty} (1 - e^{-t})$$

$$= 1 - 0 = 1$$

Limiting position is $x = 1$ m.

Question 3:

Given $a = -k/x^2$ and initial conditions $x = 1, v = 0$.

i) Prove $\frac{dx}{dt} = -\sqrt{\frac{2k(1-x)}{x}}$

$$v \frac{dv}{dx} = -\frac{k}{x^2}$$

$$v dv = -\frac{k}{x^2} dx$$

$$\int_0^v v dv = \int_1^x -\frac{k}{x^2} dx$$

$$\frac{1}{2}v^2 = \left[\frac{k}{x} \right]_1^x$$

$$\frac{1}{2}v^2 = \frac{k}{x} - k$$

$$v^2 = 2k \left(\frac{1}{x} - 1 \right) = 2k \left(\frac{1 - x}{x} \right)$$

Since the car is moving towards Woolworths (to the left), v is negative:

$$v = \frac{dx}{dt} = -\sqrt{\frac{2k(1-x)}{x}}$$

ii) **Substitution** $x = \cos^2 \theta$

$$\begin{aligned}\frac{dx}{dt} &= -\sqrt{\frac{2k(1-\cos^2 \theta)}{\cos^2 \theta}} \\ \frac{dx}{dt} &= -\sqrt{\frac{2k \sin^2 \theta}{\cos^2 \theta}} = -\sqrt{2k} \tan \theta\end{aligned}$$

From $x = \cos^2 \theta$, we have $dx = -2 \cos \theta \sin \theta d\theta$:

$$\begin{aligned}-2 \cos \theta \sin \theta \frac{d\theta}{dt} &= -\sqrt{2k} \tan \theta \\ \frac{d\theta}{dt} &= \frac{\sqrt{2k} \tan \theta}{2 \cos \theta \sin \theta} \\ \frac{d\theta}{dt} &= \frac{\sqrt{2k} \frac{\sin \theta}{\cos \theta}}{2 \cos \theta \sin \theta} = \frac{\sqrt{2k}}{2 \cos^2 \theta}\end{aligned}$$

Rearranging for dt :

$$dt = \frac{2 \cos^2 \theta}{\sqrt{2k}} d\theta = \frac{2 \cos^2 \theta}{\sqrt{2k}} d\theta$$

To find time t to reach distance D :

$$t = \int dt = \int \frac{2 \cos^2 \theta}{\sqrt{2k}} d\theta = \sqrt{\frac{2}{k}} \int \cos^2 \theta d\theta$$

Limits: $x = 1 \implies \cos^2 \theta = 1 \implies \theta = 0$.

$x = D \implies \cos^2 \theta = D \implies \theta = \cos^{-1} \sqrt{D}$.

$$t = \sqrt{\frac{2}{k}} \int_0^{\cos^{-1} \sqrt{D}} \cos^2 \theta d\theta$$

iii) **Show** $t = \sqrt{\frac{1}{2k}} (\sqrt{D - D^2} + \cos^{-1} \sqrt{D})$

Using $\cos^2 \theta = \frac{1 + \cos 2\theta}{2}$:

$$\begin{aligned}t &= \sqrt{\frac{2}{k}} \int_0^{\cos^{-1} \sqrt{D}} \frac{1 + \cos 2\theta}{2} d\theta \\ t &= 2\sqrt{\frac{2}{k}} \int_0^{\cos^{-1} \sqrt{D}} (1 + \cos 2\theta) d\theta \\ t &= \sqrt{\frac{1}{2k}} \int_0^{\cos^{-1} \sqrt{D}} (1 + \cos 2\theta) d\theta \\ t &= \frac{1}{\sqrt{2k}} [\theta + \frac{1}{2} \sin 2\theta]_0^{\cos^{-1} \sqrt{D}} \\ t &= \frac{1}{\sqrt{2k}} [\theta + \sin \theta \cos \theta]_0^{\cos^{-1} \sqrt{D}}\end{aligned}$$

Since $\cos \theta = \sqrt{D}$, then $\sin \theta = \sqrt{1-D}$.

$$t = \frac{1}{\sqrt{2k}}(\cos^{-1} \sqrt{D} + \sqrt{1-D}\sqrt{D})$$

$$t = \frac{1}{\sqrt{2k}}(\cos^{-1} \sqrt{D} + \sqrt{D-D^2})$$

Question 4:

Given $a = \frac{1}{2}(1 - \frac{1}{x^2})$ and initial conditions $x = 1, v = 2$. **Prove** $\frac{dx}{dt} = \frac{1+x}{\sqrt{x}}$

$$v \frac{dv}{dx} = \frac{1}{2}(1 - \frac{1}{x^2})$$

$$v dv = \frac{1}{2}(1 - \frac{1}{x^2}) dx$$

$$\int v dv = \int \frac{1}{2}(1 - \frac{1}{x^2}) dx$$

$$\frac{1}{2}v^2 = \frac{1}{2}(x + \frac{1}{x}) + C$$

Apply $x = 1, v = 2$:

$$\frac{1}{2}(2)^2 = \frac{1}{2}(1 + 1) + C$$

$$2 = 1 + C \implies C = 1$$

Thus, $\frac{1}{2}v^2 = \frac{1}{2}(x + \frac{1}{x}) + 1 = \frac{x^2+1+2x}{2x} = \frac{(x+1)^2}{2x}$.

$$v^2 = \frac{(x+1)^2}{x}$$

$$v = \frac{dx}{dt} = \frac{x+1}{\sqrt{x}}$$

ii) Time taken to reach $x = 3$

$$\frac{dx}{dt} = \frac{x+1}{\sqrt{x}}$$

$$\frac{\sqrt{x}}{x+1} dx = dt$$

$$\int_1^3 \frac{\sqrt{x}}{x+1} dx = \int_0^t dt$$

Let $u = \sqrt{x} \implies x = u^2 \implies dx = 2u du$:

$$\int \frac{u}{u^2+1} (2u du) = t$$

$$2 \int \frac{u^2}{u^2+1} du = t$$

$$2 \int (1 - \frac{1}{u^2+1}) du = t$$

$$2[u - \tan^{-1} u]_1^{\sqrt{3}} = t$$

Evaluating the limits:

$$t = 2(\sqrt{3} - \tan^{-1} \sqrt{3} - (1 - \tan^{-1} 1))$$

$$t = 2(\sqrt{3} - \frac{\pi}{3} - 1 + \frac{\pi}{4})$$

$$t = 2(\sqrt{3} - 1 - \frac{\pi}{12})$$

Question 5:

Given $v = \sqrt{2x}e^{-x^2}$ for $x \geq 0$. **i) Show** $a = e^{-2x^2}(1 - 4x^2)$

$$\begin{aligned} a &= v \frac{dv}{dx} \\ \frac{dv}{dx} &= \frac{d}{dx}((2x)^{1/2}e^{-x^2}) \\ \frac{dv}{dx} &= \frac{1}{2}(2x)^{-1/2}(2)e^{-x^2} + (2x)^{1/2}e^{-x^2}(-2x) \\ \frac{dv}{dx} &= \frac{1}{\sqrt{2x}}e^{-x^2} - 2x\sqrt{2x}e^{-x^2} \\ a &= (\sqrt{2x}e^{-x^2}) \left(\frac{1}{\sqrt{2x}}e^{-x^2} - 2x\sqrt{2x}e^{-x^2} \right) \\ a &= e^{-2x^2} - 2x(2x)e^{-2x^2} \\ a &= e^{-2x^2}(1 - 4x^2) \end{aligned}$$

ii) Fastest speed Fastest speed occurs when $a = 0$:

$$\begin{aligned} e^{-2x^2}(1 - 4x^2) &= 0 \\ 1 - 4x^2 &= 0 \\ x^2 &= 1/4 \implies x = 1/2 \end{aligned}$$

Substitute $x = 1/2$ into v :

$$\begin{aligned} v_{max} &= \sqrt{2(1/2)}e^{-(1/2)^2} \\ v_{max} &= 1 \cdot e^{-1/4} = e^{-0.25} \approx 0.7788 \end{aligned}$$

iii) Show $T = \int_1^2 \frac{1}{\sqrt{2x}}e^{x^2} dx$

$$\begin{aligned} v &= \frac{dx}{dt} = \sqrt{2x}e^{-x^2} \\ \frac{1}{\sqrt{2x}e^{-x^2}} dx &= dt \\ dt &= \frac{1}{\sqrt{2x}}e^{x^2} dx \end{aligned}$$

.Integrating from $x = 1$ to $x = 2$:

$$T = \int_1^2 \frac{1}{\sqrt{2x}}e^{x^2} dx$$

iv) Trapezoidal Rule Let $f(x) = \frac{1}{\sqrt{2x}}e^{x^2}$. Three function values means $n = 2$ intervals, $h = (2 - 1)/2 = 0.5$.

$$\begin{aligned}f(1) &= \frac{1}{\sqrt{2}}e^1 \approx 0.7071 \cdot 2.7183 = 1.922 \\f(1.5) &= \frac{1}{\sqrt{3}}e^{2.25} \approx 0.5773 \times 9.4877 = 5.478 \\f(2) &= \frac{1}{\sqrt{4}}e^4 = 0.5 \times 54.598 = 27.299\end{aligned}$$

Using the trapezoidal rule formula $T \approx \frac{h}{2}[f(x_0) + 2f(x_1) + f(x_2)]$:

$$\begin{aligned}T &= \frac{0.5}{2}[1.922 + 2(5.478) + 27.299] \\T &= 0.25[1.922 + 10.956 + 27.299] \\T &= 0.25[40.177] \approx 10.04\end{aligned}$$

Rounding to the nearest whole number: $T \approx 10$.

Question 6:

Given M_1 on a 30° incline, M_2 hanging vertically. M_1 accelerates down the plane at $a = 2 \text{ m/s}^2$. $g = 10 \text{ m/s}^2$. For M_1 (down the plane):

$$\begin{aligned}F_{net,1} &= M_1 g \sin 30^\circ - T \\M_1 a &= M_1 g(0.5) - T \\T &= 0.5M_1 g - M_1 a\end{aligned}$$

For M_2 (upwards):

$$\begin{aligned}F_{net,2} &= T - M_2 g \\M_2 a &= T - M_2 g \\T &= M_2 a + M_2 g = M_2(a + g)\end{aligned}$$

Equating the two expressions for T :

$$\begin{aligned}0.5M_1 g - M_1 a &= M_2(a + g) \\M_1(0.5g - a) &= M_2(a + g) \\M_1(0.5(10) - 2) &= M_2(2 + 10) \\M_1(5 - 2) &= M_2(12) \\3M_1 &= 12M_2 \\M_1 &= 4M_2\end{aligned}$$

Since $M_1 = kM_2$, then $k = 4$.

Question 7:

Given a body of weight W on a plane inclined at angle α .

Case 1: Force P parallel to the incline

To support the body, P must balance the component of weight acting down the plane:

$$P = W \sin \alpha \implies \frac{1}{P} = \frac{1}{W \sin \alpha}$$

Case 2: Force Q parallel to the horizontal

The force Q must balance the resultant of the normal force N and the weight W . The component of Q acting up the plane is $Q \cos \alpha$.

$$Q \cos \alpha = W \sin \alpha$$

$$Q = W \tan \alpha \implies \frac{1}{Q} = \frac{1}{W \tan \alpha}$$

Proof

$$\begin{aligned} \frac{1}{P^2} - \frac{1}{Q^2} &= \frac{1}{(W \sin \alpha)^2} - \frac{1}{(W \tan \alpha)^2} \\ &= \frac{1}{W^2 \sin^2 \alpha} - \frac{1}{W^2 \frac{\sin^2 \alpha}{\cos^2 \alpha}} \\ &= \frac{1}{W^2 \sin^2 \alpha} - \frac{\cos^2 \alpha}{W^2 \sin^2 \alpha} \\ &= \frac{1 - \cos^2 \alpha}{W^2 \sin^2 \alpha} \\ &= \frac{\sin^2 \alpha}{W^2 \sin^2 \alpha} \\ &= \frac{1}{W^2} \end{aligned}$$